

COURSE DESCRIPTION				
Program	Wood and Paper Technology			
Course Title	Botany Lab			
Course Code	2129	Semester	II	
Type of Course	Lab	Contact Hours	45	
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5	
CIE Marks	60	ESE Marks	40	

Introduction

Botany is the branch of biology that focuses on the study of plants, including their structure, function, growth, and classification. Through hands-on laboratory work, students can better understand how plants are adapted to their environments and how their different parts work together to sustain life. This botany lab aims to introduce basic plant structures and functions through observation and experimentation. By examining plant specimens and using simple lab tools, students develop skills in scientific observation, data collection, and analysis. The knowledge gained from this lab helps build a foundation for understanding plant biology and its importance in everyday life and the natural world. The lab also helps develop practical skills in microscopic examination and scientific analysis, which are essential for understanding and improving processes in the wood and paper industry.

Course Objectives

1	To impart knowledge on plant morphology, taxonomy and anatomy.
2	To impart knowledge on various wood defects.
3	To provide students with skills necessary to section and stain fresh plant material in preparation for study of plant anatomy.
4	To train students in the proper use of the compound light microscope and to give them experience in interpreting images that they see through the microscope in terms of how plant structure is related to function.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Morphology, Taxonomy and Anatomy			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Interpret plant diversity and types of wood defects caused by fungus	Apply
CO2	Analyse reproductive morphology, morphology of fruits and types of inflorescence	Apply
CO3	Identify plant using angiosperm taxonomy	Apply
CO4	Analyse stem, root, leaf and wood anatomy	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	2	-	-	1	-	1
CO2	3	2	-	-	2	1	-	-	1	-	1
CO3	3	1	-	-	1	-	-	-	-	-	1
CO4	3	3	-	2	3	2	-	-	1	-	1
Total	12	8	-	2	8	5	-	-	3	-	4
Correlation Level	3	2	-	2	2	1.67	-	-	1	-	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
							CIE	ESE	Total	CIE	ESE	Total
L	T	P	S									

0	0	3	0	1.5	45	1.5				60	40	100	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Plant diversity	Microscopic observation and Identification of algae(Spirogyra), Fungi(Rhizopus), Bryophyta(Moss), Pteridophyta(Nephrolepis)	CO1	Apply	3
2	Wood defects caused by fungi	Identification of fungal wood defects - Rots, Stains etc.	CO1	Apply	3
Module II					
3	Morphological study	Identification of Modification of stem, root and leaf	CO2	Apply	3
4	Inflorescence	Identification of types of inflorescence	CO2	Apply	2
5	Fruit	Identification of types of fruit	CO2	Apply	2
	Series Lab Exam - I		CO2	Apply	3
Module III					
6	Plant taxonomy	Conduct study of vegetative and floral characters of any one member each of the following families Annonaceae, Euphorbiaceae, Verbenaceae, Rutaceae, Fabaceae, Myrtaceae, Poaceae	CO3	Apply	11
Module IV					
7	Cell and tissues	Demonstrate plant cell - Onion peel Demonstrate plant tissues - C.S of Eupatorium or Cucurbit stem	CO4	Apply	2
8	Primary structure	Examine primary structure of Dicot and Monocot Stem, Root and Leaf	CO4	Apply	6
9	Secondary structure	Identify secondary thickening in Dicot stem- Vernonia or Eupatorium	CO4	Apply	2
10	Wood anatomy	Demonstrate Heart wood and Sap wood, Hard wood and Soft wood, Tyloses, Growth rings	CO4	Apply	2
11	Wood defects	Identify Wood defects - knots, shakes, warping, grain issues, insect damage etc	CO4	Apply	3
	Series Lab Exam - II		CO4	Apply	

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Observe plants around your surroundings and classify them into major plant groups such as Algae, Bryophyte, Pteridophyte, Gymnosperm and Angiosperm	1.5
2	Observe mold on moist bread under microscope	1.5

3	Observe wooden items around your surroundings for fungal wood defects. Collect photographs of fungal wood defects	1.5
4	Interpreting the theory concepts and finishing lab records	1.5
Module II		
5	Observe and identify common stem, root and leaf modifications	1.5
6	Collect common flowers and identify the type of inflorescence. Collect common fruits and identify the type of fruit.	1.5
7	Collect flowers and identify their parts	1.5
8	Interpreting the theory concepts and finishing lab records	1.5
Module III		
9	Observe and identify plants belonging to common families	1.5
10	Practice writing scientific names using binomial nomenclature	1.5
11	Practice drawing floral diagram and floral formula	1.5
12	Interpreting the theory concepts and finishing lab records	1.5
Module IV		
13	Observe permanent slides of root, stem, and leaf primary structure under a microscope. Record observations from microscopic slides	1.5
14	Identify growth rings and estimate age of a tree. Observe permanent slides of major woods under a microscope. Observe wood defects like knots, insect damage, shakes etc.	1.5
15	Collect photographs of wood defects. Interpreting the theory concepts and finishing lab records	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Practical botany	A.M Bendre	Rastogi publications
2	Practical botany	Dr. Rekha Arya	Hindisahithyasadan

REFERENCE			
Sl. No.	Title	Author	Publication
1	Botany lab manual	Dr. S Anil Kumar	New jyothi publications

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://ncert.nic.in/science-laboratory-manual.php	All
2	https://www.scribd.com/document/822955823/Botany-Practical-Manual-2024-25-Copy	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Compound Microscope	Eye piece : 10x Objective lens : 4x,10x	10
2	Dissecting Microscope	Eye piece : 10x Objective : 1x/2x	10
3	Glass slides	75x25mm	30
4	Cover slips	18x18 mm or 22x22mm	30
5	Safranin stain		1 bottle (125ml)

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15		
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10

Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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